COURSE SYLLABUS1

PHY 400, PHY 880, *Lasers and Optomechanics*
Physics Building 208, Monday-Wednesday 12:45-2:05 pm

Instructor	Professor Craig Cahillane	Phone	202 907 9680
Office	PHY 257	E-mail	ccahil01@syr.edu
Office Hours	Thursday 2:30 - 3:30 pm		

Course Website: https://ccahilla.github.io/lasers-and-optomechanics/

Course Description:

Lasers and optomechanical devices are permeating the scientific and engineering landscape more and more, being used in quantum computing devices, gravitational-wave interferometers, fusion energy pulsed drivers and diagnostics, and more.
Understanding lasers and interferometers from a mathematical perspective is a highly important and nontrivial endeavour.
Lasers and Optomechanics seeks to lay the mathematical framework to understand laser interferometers from basic principles.
The successful student will be well-positioned to understand how any interferometer works, and establish a strong understanding of a laser and how it operates.
The end of the course features a final project where students apply their new knowledge to a major scientific facility or important laser technology or concept, and give a 30 minute presentation on any laser-related topic of their choosing.

Prerequisite / Co-requisite:
PHY 360 (Vibrations, Waves and Thermo) or equivalent
PHY 524 (Electromagnetics I) or equivalent

Audience:
Physicists and Optical Engineers
Senior Undergraduates and Graduate Students

Credits:
3 Credits

Course Fees and/or Costs:
None

Learning Objectives:
After taking this course, students will be able to:

Understand the basics of lasers, how they function, what are their modes of operation, and how to build a laser.
Understand and calculate the optomechanical responses of complex interferometers.
Understand Pound-Drever-Hall laser frequency stabilization, a crucial technique for controlling basic interferometers.
How lasers and interferometers are useful, what kinds of problems can they be used to solve.
Research and present on current state-of-the-art facilities and laser concepts pushing scientific progress in the field.

Shared Competencies Course Tags [UNDERGRADUATE ONLY]
Scientific Inquiry and Research Skills

Required Texts / Supplies:
None, the course website has Chapters to follow at
https://ccahilla.github.io/lasers-and-optomechanics/

Recommended Text: Siegman *Lasers*
Optica Free electronic publication of Lasers:
https://opg.optica.org/content/bookshelf/book/item/lasers

Course Requirements and Expectations:

Students will be expected to know python, jupyter notebooks, and git, as all homework assignments will be turned in by submitting a jupyter notebook to git repository the student has created and shared with the professor.

Students will be expected to have working knowledge of the following

1. Waves
2. Classical Mechanics
3. Complex Numbers
4. Differential Equations
5. Laplace Transforms
6. `python`
7. Linear Algebra

We will review topics as necessary, but previous exposure will be useful.

The Spring 2026 course features around four in-depth homework assignments, five in-class quizzes, four in-class tutorials done amongst the students and with the professor.

The professor lectures early in the semester and lays some mathematical groundwork, interspersed with the quizzes.
After Spring Break, the in-class tutorials will be given, where students will solve complex interferometer configurations with the help of the professor.
Then students will take over class time with their 30-minute final project presentations, each student issuing peer reviews.

Grading:

We'll have four major course components:

1. In-class Work
2. Homework
3. In-class Quizzes (every Wednesday)
4. End-of-year Special Topics Review Project by each student

The grade breakdown is as follows:

Component	**Percent of Final Grade**
In-class Work	15%
Homework	25%
Quizzes	25%
Review Project	35%

In-class Work is the in-class tutorials and peer-assessments of the final projects. These will be graded for completion and effort.

Homeworks will be due every two-three weeks on Monday. I will release them on the course website in Jupyter Notebooks with questions that must be completed.

All homeworks must be turned in via Jupyter Notebook. I recommend downloading the homework file from the gitlab: https://github.com/ccahilla/lasers-and-optomechanics

Most problems will require math to be done. These may be done on paper and scanned in, or directly in LaTeX.

Some HW problems will require plots to be made. I recommend doing those in python in the notebook.

Quizzes will be a short in-class question reviewing the previous week's physics concept from lecture, readings, and homework.

The **End-of-year Review Project** will be to prepare a review in a Jupyter Notebook like the ones in this class about a topic in interferometry, lasers, optomechanics, or anything else related to the course.

Grading Table:

Grade	Grade Point/credit	Percentage Range	Total Points
A	4.000	100-93	
A-	3.667	93-88	
B+	3.333	88-84	
B	3.000	84-80	
B-	2.667	80-76	
C+	2.333	76-72	
C	2.000	72-68	
C-	1.667	68-64	
D	1.000	60-50	
F	0	50-0	

[Source: Academic Rules Section 8.4 Grades

*__Graduate courses may not use grades D or D-.__ Only Law courses may use the D- grade. Remember to adjust grade points, percentages, and total points accordingly.

Course Specific Policies:

Homeworks may be done together with other students, but each student must submit their own assignment done by their own hand.

Quizzes are in-class and in-person, no exceptions.

In-class Tutorials and Peer Assessments will be graded for attendance and effort.

Final Projects will be jupyter notebooks in the same style as the lectures, complete with motivation, technical details, illustrating images and plots, and references.
A rubric will be presented in class so students know what they will be graded on.

Course Schedule:

Spring 2026 Lasers and Optomechanics Class Schedule				
Class	**Date**	**Lecture Topic**	**Homework**	**Quiz**
1	Monday, Jan 12	Introduction and Complex Number Review 1		
2	Wednesday, Jan 14	Complex Number Review 2		Complex Numbers Review
	Monday, Jan 19	*Martin Luther King Jr. Day*		
3	Wednesday, Jan 21	Electromagnetic Radiation 1		
4	Monday, Jan 26	Electromagnetic Radiation 2	Plane Waves and Math Review - Homework 01 Due	
5	Wednesday, Jan 28	Electromagnetic Radiation 3		Double Slit Interference
6	Monday, Feb 2	Intro to Lasers 1		
7	Wednesday, Feb 4	Intro to Lasers 2		
8	Monday, Feb 9	Fabry-Perot Interferometers 1		
9	Wednesday, Feb 11	Fabry-Perot Interferometers 2		Atomic Rates
10	Monday, Feb 16	Fabry-Perot Interferometers 3	Lasers and Fabry Perots - Homework 02 Due	
11	Wednesday, Feb 18	Michelson Interferometers 1		Fabry-Perot Plots
12	Monday, Feb 23	Michelson Interferometers 2		
13	Wednesday, Feb 25	Michelson Interferometers 3		Mach Zender as Intensity Controller
14	Monday, Mar 2	Modulations 1		
15	Wednesday, Mar 4	Modulations 2		
	Monday, Mar 9	*Spring Break*	Michelsons, Modulations, and Transfer Functions - Homework 03 Due	
	Wednesday, Mar 11	*Spring Break*		
16	Monday, Mar 16	Dither Locking (In-class Tutorial)		
17	Wednesday, Mar 18	Pound-Drever-Hall (In-class Tutorial)		

18	Monday, Mar 23	Coupled Cavities 1 (In-class Tutorial)		
19	Wednesday, Mar 25	Coupled Cavities 2 (In-class Tutorial)		
20	Monday, Mar 30	Shot Noise Lecture		
21	Wednesday, Apr 1	Student Presentations (Nico and Mason)		
22	Monday, Apr 6	Student Presentations (Julia and Carter Halpin)	Transmission Dither Locking, PDH Michelson, and FPMI - Homework 04 Due	
23	Wednesday, Apr 8	Craig gone at ARPA-E Summit		
24	Monday, Apr 13	Student Presentations (Jared and Breck)		
25	Wednesday, Apr 15	Student Presentations (Alex Corriea and Carter Thompson)		
26	Monday, Apr 20	Student Presentations (Jandrie and Caroline)		
27	Wednesday, Apr 22	Student Presentations (Shreyan and Ella)		
28	Monday, Apr 27	Student Presentations (Eadin)		

SU Policies and Services: